Lecture:-1

Module -1

Topic: Introduction to DS : DATA , INFORMATION, DATA STRUCURE, ALGORITHM, ANALYSIS OF ALGORITHM,TYPES OF DATA STRUCTURE

Basic Terminologies :-

- Data:- Raw facts and figures without context.
  Ex- "23", "apple" , "5.6".
- Information:- Processed or organized data that is meaningful.
  Ex- "Student scored 95 marks"
- Data processing:- The operation on data to retrieve, transform and classify it.
  Ex- Storing student marks.

1. What is data structure?

➤ The way of organizing and sorting data so that it can be it can be accessed and modified efficiently.

Note:-

Data Structure Operations:

- Insertion
- Deletion
- Traversal
- Searching
- Sorting

2. What is insertion operation?

➤ The process of adding a new element into a data structure.

3. What is deletion operation?

➤ The process of removing an element from a data structure.

4. What is traversal operation in data structures?

➤ The process of visiting each element of data structure exactly once.

5. Traversal means accessing each element of a data structure. ➤ True.

6. What is meant by elementary data organization ?

➤ The basic method of arranging data items in memory for easy processing.

7. Define algorithm.

➤ A finite set of step-by-step instructions used to solve a problem.

8. An algorithm must have a finite number of steps.

➤ True.

9. What is algorithm analysis?

➤ The process of evaluating the performance of an algorithm in terms of time and space.

Types of data structure:-

1. Primitive Data Structures :-

These are the basic data types provided by a programming language.

- Integer
- Float
- Character
- Boolean
- Pointer

2. Non-Primitive Data Structure :-

A non-primitive data structure is a complex data structure built from primitive data types that is used to store and organize multiple data elements efficiently.

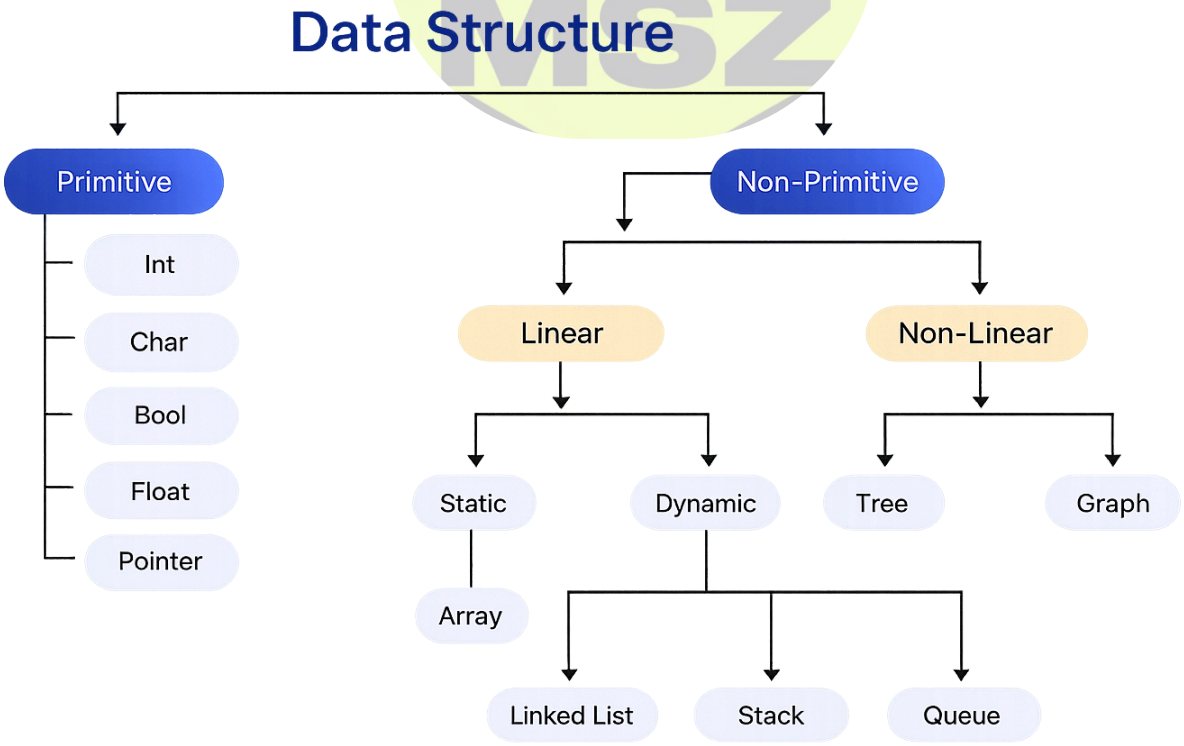
It is classified to two types ⟶

- **Linear Data Structure** - Linear data structures organize data in a sequential manner where each element is adjacent to the previous one.

  Ex:- Array, Linked List, Stack, Queue

- **Non-Linear Data Structure** - Non-linear data structures organize data hierarchically, connecting elements in a more complex, branching manner.

  Ex:- Tree, Graph



Long Question (5 & 15 Marks)

1. Explain linear and non-linear data structure with example.

➤ Linear Data Structure: A linear data structure is a type of data structure in which elements are arranged sequentially (one after another). Each element is connected to its previous and next element.

Characteristics:

- Elements are stored in a linear or sequential order.
- Each element (except the first and last) has one predecessor and one successor.

Examples:

1. Array - Elements are stored in contiguous memory locations.

Example: 10, 20, 30, 40

2. Linked List- Each element (node) points to the next node.

3. Stack- Follours LIFO (Last in First out) principle.

Example: Stack of plates

4. Queue- Folloues FIFO (First In First Out) principle.

Example: Queue of people waiting for a bus.

➤ Non-Linear Data Structure: A non- linear data structure is a structure where elements are not arranged sequentially. Data elements are connected in a hierarchical or network form.

Characteristics:

- Elements are not stored in a single sequence.
- One element can be connected to multiple elements.
- Traversal follows different paths.

Examples:

1.Tree- A hierarchical structure consisting of nodes and branches.

2. Graph- A structure consisting of vertices (nodes) and edges.

2. What are the difference between linear and non-linear data structures?

➤

Basis	Linear Data Structure	Non-Linear Data Structure
1) Arrangement of Data	Elements are arranged in a sequential order.	Elements are arranged in a hierarchical or network form.
2) Relationship Between Elements	Each element has one predecessor and one successor (except first and last).	One element can be connected to multiple elements.
3) Traversal	Data elements are traversed in a single straight path.	Data elements are traversed through multiple paths.
4) Structure	Forms a single level structure.	Forms a multilevel structure.
5) Implementation	Simpler to implement.	More complex to implement.
6) Memory Utilization	Memory is usually sequentially organized.	Memory is not necessarily sequentially organized.
7) Examples	Array, Linked List, Stack, Queue	Tree, Graph

3. Define algorithm? What are the measures of performance of an algorithm ? Explain it. What are the characteristics of algorithm?

➤ Algorithm: An algorithm is a finite set of step-by-step instructions used to solve a problem or perform a specific task.

It takes input, processes it through a series of steps, and produces an output.

Example: Algorithm to add two numbers

1. Start
2. Input two numbers A and B
3. Compute Sum = A+B
4. Display Sum

➤ Measures of Performance of an Algorithm:

The performance of an algorithm is mainly measured by two factors:

1.Time Complexity:

- It measures the amount of time required by an algorithm to execute. It depends on the size of the input (n).

Example: Linear search has time complexity O(n).

2. Space Complexity:

- It measures the amount of memory required by an algorithm during execution. Includes memory for variables, data structures, and program instructions.

Example: Linear search has space complexity O(1).

➤ Characteristics of Algorithm:

1. Input:- An algorithm should have zero or more inputs.
2. Output:- It must produce at least one output.
3. Definiteness:- Each step of the algorithm must be clear and unambiguous.
4. Finiteness:- The algorithm must terminate after a finite number of steps.
5. Effectiveness:- Each step should be simple and executable within a finite time.
6. Generality:- The algorithm should be applicable to a class of problems, not just a single instance.

